

# Foundations of Operations Research

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# Chapter 1: Introduction

## What is Operations Research?

O.R. is a branch of applied mathematics in which **mathematical models** and **quantitative methods** (e.g., optimization, game theory, simulation) are used to analyze **complex decision-making problems** and find (near-)optimal solutions.

**Overall goal:** To help make better decisions

Interdisciplinary field at the interface of applied mathematics, computer science, economics and industrial engineering.

# 1.1 Decision-making problems

Problems in which we must choose a (feasible) solution among a large number of alternatives based on one or several criteria.

## Examples

- 1 Assignment problem
- 2 Network design
- 3 Shortest paths
- 4 Personnel scheduling
- 5 Service management
- 6 Multicriteria problem

Complex decision-making problems that are tackled via a **mathematical modelling approach** (mathematical models, algorithms and computer implementations).

## 1. Assignment problem

Given  $m$  jobs and  $m$  machines, suppose that each job can be executed by any machine and that  $t_{ij}$  is the execution time of job  $J_i$  on machine  $M_j$ :

|       | $M_1$ | $M_2$ | $M_3$ |
|-------|-------|-------|-------|
| $J_1$ | 2     | 6     | 3     |
| $J_2$ | 8     | 4     | 9     |
| $J_3$ | 5     | 7     | 8     |

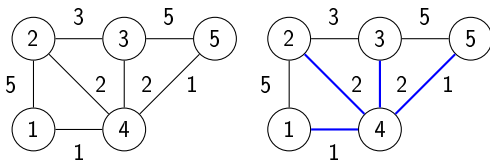
Decide which job to assign to each machine so as to minimize the total execution time. Each job must be assigned to exactly one machine, and each machine to exactly one job.

Number of feasible solutions?  $m!$  possible assignments

## 2. Network design

Decide how to connect  $n$  cities (offices) via a collection of possible links so as to minimize the total link cost.

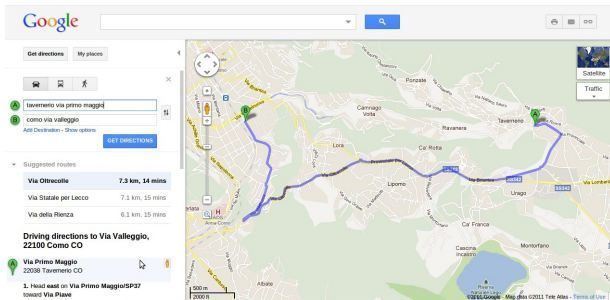
Given a graph  $G = (N, E)$  with a node  $i \in N$  for each city and an edge  $\{i, j\} \in E$  of cost  $c_{ij}$  for each link, select a subset of edges of minimum total cost, guaranteeing that all pairs of nodes are connected.



Number of alternative solutions? At most  $2^m$ , where  $m = |E|$ .

### 3. Shortest path

Given a direct graph that represents a road network with distances (traveling times) for each arc, determine the shortest (fastest) path between two points (nodes).



#### 4. Personnel scheduling:

Determine the week schedule for the hospital personnel so as to minimize the number of people involved (physicians, nurses, ...) while meeting the daily requirements.

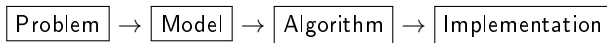
#### 5. Service management:

Determine how many counters/desks to open at a given time of the day so that the average customer waiting time does not exceed a certain value (guarantee a given service quality).

#### 6. Multicriteria problem:

Decide which laptop to buy considering the price, the weight and the performance.

## 1.2 Scheme of an O.R. study



Main steps:

- define the problem
- build the model
- select or develop an appropriate algorithm
- implement or use an efficient computer program
- analyze the results

with feedbacks (the previous steps are reconsidered whenever useful).

A mathematical **model** is a simplified representation of a real-world problem.

To define it we must identify the fundamental elements of the problem and the main relationships among them.

## 1.3 O.R. history

Origin in WW II: teams of scientists were asked to do **research** on the most efficient way to conduct the **operations** (optimize allocation of scarce resources).

Examples: radar location, manage convoy operations, logistics, ...

After the war, the techniques became public and began to be applied to problems in business, industry and society.

During the industrial boom, the substantial increase in the size of the companies/organizations gave rise to more complex decision-making problems.

Favourable circumstances:

- Fast progress in O.R. and in Numerical Analysis methodologies
- Advent and diffusion of computers (computing power and software).

O.R. and Management Science are often used as synonyms

## Example 1. Production planning

A company produces 3 types of electronic devices:  $D_1$ ,  $D_2$ ,  $D_3$

Main phases of the production process: assembly, refinement and quality control.

Time (in minutes) required for each phase and product:

|                 | $D_1$ | $D_2$ | $D_3$ |
|-----------------|-------|-------|-------|
| Assembly        | 80    | 70    | 120   |
| Refinement      | 70    | 90    | 20    |
| Quality control | 40    | 30    | 20    |

Available resources within the planning horizon (depend on workforce) in minutes:

| Assembly | Refinement | Quality control |
|----------|------------|-----------------|
| 30 000   | 25 000     | 18 000          |

Unitary profit (in KEuro):

| $D_1$ | $D_2$ | $D_3$ |
|-------|-------|-------|
| 1.6   | 1     | 2     |

Assumption: the company can sell whatever it produces.

Give a **mathematical model** for determining a production plan which maximizes the total profit.



## Example 2. Portfolio selection problem

An insurance company must decide which investments to select out of a given set of possible assets (stocks, bonds, options, gold certificates, real estate, ...).

| Investments                   | area    | capital ( $c_j$ K€) | expected return ( $r_j$ ) |
|-------------------------------|---------|---------------------|---------------------------|
| A (automotive)                | Germany | 150                 | 11%                       |
| B (automotive)                | Italy   | 150                 | 9%                        |
| C (ICT)                       | U.S.A.  | 60                  | 13%                       |
| D (ICT)                       | Italy   | 100                 | 10%                       |
| E (real estate)               | Italy   | 125                 | 8%                        |
| F (real estate)               | France  | 100                 | 7%                        |
| G (short term treasury bonds) | Italy   | 50                  | 3%                        |
| H (long term treasury bonds)  | UK      | 80                  | 5%                        |

Available capital = 600 KEuro

At most 5 investments to avoid excessive fragmentation.

Geographic diversification to limit risk:  $\leq 3$  investments in Italy and  $\leq 3$  abroad.

Give a **mathematical model** for deciding which investments to select so as to maximize the expected return while satisfying the constraints.

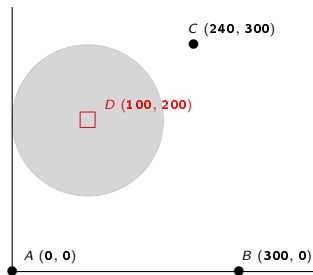


**Variant:** In order to limit the risk, if any of the ICT investment is selected then at least one of the treasury bond must be selected.

**Model:**

### Example 3. Facility location

Consider 3 oil pits, located in  $A$ ,  $B$  and  $C$ , from which oil is extracted.



Connect them to a refinery with pipelines whose cost is proportional to the square of their length.

The refinery must be at least 100 km away from point  $D$  but the oil pipelines can cross the corresponding forbidden zone.

Give a **mathematical model** to decide where to locate the refinery so as to minimize the total pipeline cost.



## 1.5 Decision-making problems

- Number of decision makers (who decides?)
- Number of objectives (based on which criteria?)
- Level of uncertainty in the parameters (based on which information?)

One decision maker, one objective  $\Rightarrow$  Mathematical programming

One decision maker, several objectives  $\Rightarrow$  Multi-objective programming

Uncertainty level  $> 0 \Rightarrow$  Stochastic programming

Several decision makers  $\Rightarrow$  Game theory

## 1.6 Mathematical Programming/Optimization (MP)

Decision-making problems with a **single decision maker**, a **single objective** and reliable parameter estimates:

$$\min f(x) \quad \text{with } x \in X \quad (1)$$

- **decision variables**  $x \in \mathbb{R}^n$  : numerical variables whose values identify a solution of the problem
- **feasible region**  $X \subseteq \mathbb{R}^n$  distinguishes between feasible and infeasible solutions (via constraints)

$$X = \left\{ x \in \mathbb{R}^n : g_i(x) \begin{cases} = \\ \leq \\ \geq \end{cases} 0, i = 1, \dots, m \right\}$$

- **objective function**  $f : X \rightarrow \mathbb{R}$  expresses in quantitative terms the value or cost of each feasible solution.

Observation:  $\max\{f(x) : x \in X\} = -\min\{-f(x) : x \in X\}$

# Global optima

Solving a mathematical programming (MP) problem consists in finding a **global minimum**, i.e., a vector  $x^* \in X$  such that

$$f(x^*) \leq f(x) \quad \forall x \in X$$

It may happen that the problem

- ❶ is infeasible ( $X = \emptyset$ ),
- ❷ is unbounded ( $\forall c \in \mathbb{R}, \exists x_c \in X$  such that  $f(x_c) \leq c$  or  $f(x_c) \geq c$ ),
- ❸ has a single optimal solution,
- ❹ has a large number (even infinite) of optimal solutions.

# Local optima

When the problem is very hard we must settle for a feasible solution that is a **local minimum**, i.e., a vector  $\hat{x} \in X$  such that

$$f(\hat{x}) \leq f(x) \quad \forall x \in X \text{ with } \|x - \hat{x}\| \leq \epsilon$$

for an appropriate value  $\epsilon > 0$ .

An optimization problem can have many local optima.

# Special cases of Mathematical Programming

- **Linear Programming (LP)**

$$f \text{ linear and } X = \left\{ \mathbf{x} \in \mathbb{R}^n : g_i(\mathbf{x}) \begin{cases} = \\ \leq \\ \geq \end{cases} 0, i = 1, \dots, m \right\} \text{ with } g_i \text{ linear } \forall i$$

Example: Production planning

- **Integer Linear Programming (ILP)**

$$f \text{ linear and } X = \left\{ \mathbf{x} \in \mathbb{R}^n : g_i(\mathbf{x}) \begin{cases} = \\ \leq \\ \geq \end{cases} 0, i = 1, \dots, m \right\} \cap \mathbb{Z}^n \text{ with } g_i \text{ linear } \forall i$$

Example: Portfolio selection

- **Nonlinear Programming (NLP)**

$$f \text{ (non) convex/regular and } X = \left\{ \mathbf{x} \in \mathbb{R}^n : g_i(\mathbf{x}) \begin{cases} = \\ \leq \\ \geq \end{cases} 0, i = 1, \dots, m \right\} \text{ with } g_i \\ \text{(non) convex/regular } \forall i$$

Example: Facility location (with  $g_i$  convex)

# Historical sketch of Mathematical Programming

1826/1827: **Joseph Fourier** presents a method to solve systems of linear inequalities (Fourier-Motzkin) and discusses some LPs with 2-3 variables.

1939: **Leonid Kantorovitch** lays the bases of LP (Nobel prize 1975).

1947: **George Dantzig** proposes independently LP and invents the simplex algorithm.

1958: **Ralph Gomory** proposes a cutting plane method for ILP problems.



J. Fourier  
(1786-1830)



L. Kantorovitch  
(1912-1986)



G. Dantzig  
(1914-2005)



R. Gomory  
(1929-)

## 1.6 Multi-objective programming

Multiple objectives can be taken into account in different ways

Suppose we wish to minimize  $f_1(\mathbf{x})$  and maximize  $f_2(\mathbf{x})$  (laptop example:  $f_1$  is cost and  $f_2$  is performance)

- 1 Turn it into a **single objective problem** by expressing the two objectives in terms of the same unit (e.g., monetary unit)

$$\min \lambda_1 f_1(\mathbf{x}) - \lambda_2 f_2(\mathbf{x})$$

for appropriate scalars  $\lambda_1$  and  $\lambda_2$ .

- 2 Optimize the **primary objective** function and turn the other objective into a constraint:

$$\max_{\mathbf{x} \in \tilde{X}} f_2(\mathbf{x}) \quad \text{where } \tilde{X} = \{\mathbf{x} \in X : f_1(\mathbf{x}) \leq c\}$$

for an appropriate constant  $c$ .

## 1.7 Mathematical programming versus simulation

Both approaches involve constructing a **model** and designing an **algorithm**.

### Mathematical programming

Problem can be “well” formalized

Algorithm yields a(n optimal) solution

The results are “certain”

Example: assignment

### Simulation

Problem is difficult to formalize

Algorithm simulates the evolution of the real system and allows to evaluate the performance of alternative solutions.

The results need to be interpreted

Example: service counters

## 1.8 The impact of Operations Research

| company                          | sector                                                      | results                              |
|----------------------------------|-------------------------------------------------------------|--------------------------------------|
| Taco Bell<br>(fast food)         | personnel scheduling                                        | 7.6M\$ annual savings                |
| American Airlines                | design fare structure, overbooking and flights coordination | + 500M\$                             |
| Harris Corp.<br>(semiconductors) | production planning                                         | 50% $\Rightarrow$ 95% orders on time |
| GM - car rental                  | use of car park                                             | +50M\$ per year avoided bankruptcy   |
| HP - printers                    | modify production line                                      | doubled production                   |
| Bosques Arauco                   | harvesting logistics and transport                          | 5M\$ annual savings                  |
| IBM                              | supply chain re-engineering                                 | 750M\$ annual savings                |

Most of the top managers interviewed by “Fortune 500” declared that they used O.R. methodologies.

## O.R. & Analytics Success Stories

SHARE:



JOIN INFORMS

RENEW MY INFORMS MEMBERSHIP



Faster and Further

Improving train travel mile by mile



Transforming Aviation  
Security



Increasing Bike-share  
Efficiency



<https://www.informs.org/Impact/O.R.-Analytics-Success-Stories>

Link to Analytics for a Better World

- Significant impact not only for large companies and organizations.
- In rapidly evolving contexts, strongly competitive, highly complex and uncertain, it is crucial to identify and implement efficient solutions.
- The huge amount of data available with modern information systems (Big data) opens new avenues. . .